(mean $\rho = 0.373$), followed by other combinations and individual layers in the low-to-mid range. This is plotted in Figure 1 for compound compositionality scores; head and modifier scores follow the same trends (see Appendix C). The traditionally used combination of the last four hidden layers (9–12) is not among the best ones. It is in fact the single weakest in terms of maximum values: 0.248 for compound scores, close to 0.5 below the best implementation. However, it also has a comparatively high minimum value (-0.262), suggesting it is more robust to the effect of other parameters.

Compositionality estimates. We summarize the impact of compositionality estimates by looking at performance across the five types of embeddings which constitute the basis of subsequent score calculations (Table 7); for a summary of results on individual estimates, see Appendix D.

	modif	head	comp	cont	cls
COMP	0.135	0.274	0.245	0.172	-0.128
	-0.383	-0.133	-0.324	-0.649	-0.649
	0.615	0.630	0.666	0.706	0.611
HEAD	0.071	0.242	0.194	0.130	-0.161
	-0.384	-0.130	-0.327	-0.606	-0.606
	0.464	0.645	0.598	0.645	0.558
MODIF	0.106	0.167	0.164	0.133	-0.094
	-0.274	-0.130	-0.229	-0.476	-0.476
	0.553	0.415	0.517	0.553	0.477

Table 7: Spearman's ρ (mean, min, max) for embedding types, across all direct and composite estimates if used.

Looking at the mean values, `head` performs stronger than `modif` as well as `comp` across the prediction targets. This is coherent with the dominant role of the head in the morphological constituency of compounds. However, the maximum values out of the three are obtained by `comp` for the compound-level score, `head` for the head score, and `modif` for the modifier score. This indicates that representations targeting the whole compound or a constituent of interest are successful in capturing information specific to the respective element.

As for the two other embedding types, the mean values for `cont` follow `head` and `comp` across prediction targets. But its maximum values are the single best (compound-level prediction) or joint best (head and modifier predictions). By contrast, `cls` is clearly in the lower range of performance. Its maximum values remain around 0.1 points behind the best implementations. This shows that using the linguistic context surrounding the compound – modeled by `cont` – is beneficial, and clearly more so than using a representation of the full sequence.[4]

This might be explained by the redundancy of using a representation of a compound or its constituent, and comparing it with a representation of the full sequence which encodes the same element. It could also be the case that, as suggested before, token-level embeddings are strongly influenced by their immediate linguistic context; a balanced representation of a broader range of information might be complementary. Whatever the case, the results show that the similarity between a compound and its linguistic context is reflective of the degree of compositionality. This might explain why previous implementations using neutral contexts were less successful in capturing these trends.

5.2 Ablation study

In order to further validate the parameter-level trends observed thus far, we conduct an ablation study. We start from the parameter constellation that obtained the best results on predicting the compositionality of the compound as a whole (see Table 2; the top two configurations obtained identical results). For each parameter, we then test all other potential values, one at a time, while keeping the remaining parameters unchanged. The results are presented in Figure 2.

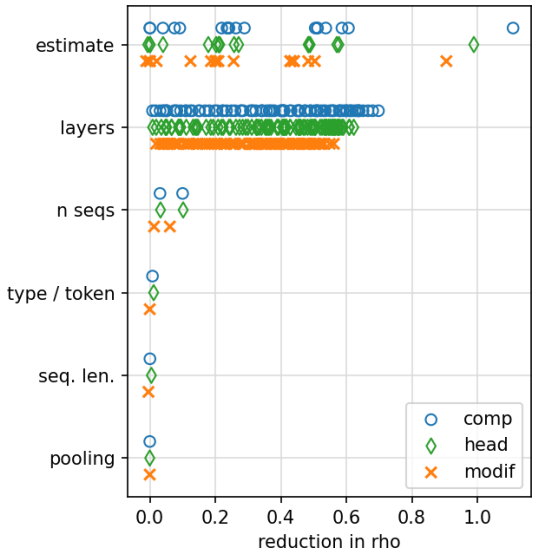

Figure 2: Effect of alternative parameter values compared to the top parameter constellation.

[4] A more direct comparison would involve a merged representation of all the tokens in the sequence rather than the `cls` embedding. We nevertheless think that we capture the same range of information.

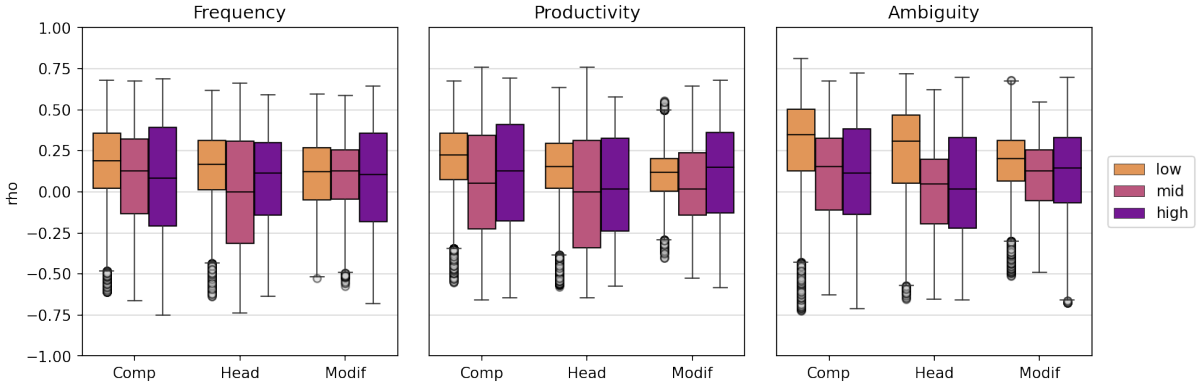

Figure 3: Effect of empirical properties of the head on model performance, observed across the evaluated implementations. Values on the x-axis indicate prediction targets (compound, head, and modifier scores).

Several parameters exhibit no or limited differences with respect to the top parameter constellation: pooling functions, minimum sequence length, and type vs. token-level modeling. Clearer effects on performance are observed for the number of modeled sequences; the drop is strongest for the smallest number of sequences. The choice of layers shows a very strong effect, with a clear tendency towards a drop in performance with both (i) a shift from initial to later hidden states, and (ii) an increase in layer span size. The strongest effect is exhibited by the estimate used to predict the degree of compositionality. All weakest estimates (reduction in $\rho > 0.4$) involve the `cls` embedding. All strongest estimates (reduction in $\rho < 0.1$) involve the `cont` embedding.

These results overall confirm the trends previously discussed for individual parameters. They also provide further evidence that model performance is most strongly affected by the choice of representational information to be used, i.e. the layers and the modeled tokens in the sequence. Suboptimal values for either of these parameters can lead to compositionality predictions that are fully decorrelated from human judgment. Performance does not depend to the same extent on the way in which data is preprocessed and representational information is combined. However, the effect of some of these parameters is not negligible, especially cumulatively, and as such should not be disregarded.

5.3 Empirical properties of compounds

We now turn to the potential impact of empirical properties of the compounds on model performance. We focus on key characteristics of the compounds' heads, given their importance indicated by the trends for compositionality estimates.

As previously stated, we examine the impact of three empirical properties: frequency, productivity, and ambiguity (for sources of this information, see Section 3.3). For each property, we rank the 280 compounds based on the values they exhibit and split them into five sets containing 56 compounds each. We retain the first, third, and fifth set, which we take to clearly reflect the low, mid, and high ranges for each empirical property. The remaining two sets are excluded in order to avoid overlapping or closely similar values in adjoining sets.

A summary of the splits across the three features is presented in Table 8. For each feature, we compute correlations with human judgments separately for each of the three splits of compounds, across all evaluated constellations of parameters. The distribution of the obtained values is plotted in Figure 3 and further discussed below.

Feature	Mean	Std.	Example
Frequency	42	± 30	*silver* ***spoon***
(thousands)	452	± 108	*labor* ***union***
	3,614	± 2,438	*crash* ***course***
Productivity	7	± 5	*night* ***owl***
	75	± 19	*time* ***difference***
	448	± 208	*birth* ***rate***
Ambiguity	2	± 1	*research* ***project***
	5	± 1	*flea* ***market***
	13	± 4	*application* ***form***

Table 8: Mean and standard deviation for the low, mid, and high range splits across empirical features. A sample compound from each split is provided.

Frequency. In predicting the compound and head compositionality scores, the best performance is obtained for low-frequency heads (mean $\rho = 0.18$

and 0.16, respectively). For the compound score, performance across the mid and high range is similar; for the head score, it is clearly lower in the mid range. As for the modifier score, performance is overall stable (mean $\rho = 0.10$ across the three splits); this is not especially surprising given the focus on the frequency of the head. This contrasts the previously reported improvements in performance with a higher number of modeled sequences. However, head frequency is correlated with both productivity and ambiguity ($\rho = 0.87$ and 0.50, respectively, across the 280 compounds). Poorer performance in higher frequency ranges is consistent with an indirect effect of these two properties.

Productivity. Overall, low-productivity heads clearly obtain the best results for compound and head compositionality scores (mean $\rho = 0.21$ and 0.16, respectively). Low- and high-productivity heads obtain similar results for the modifier scores (mean $\rho = 0.11$ and 0.12, respectively). Across the prediction targets, mid-productivity heads have the poorest performance; this drop is the strongest for head scores (mean $\rho = 0.01$). The overall better performance at modeling low-productivity heads is likely explained by their very nature of being used with fewer distinct modifiers, which might facilitate the learning of those compound meanings. This is opposed to higher productivity ranges, which potentially imply more dispersion.

Ambiguity. Across the prediction targets, low-ambiguity heads clearly have the strongest performance (mean $\rho = 0.30$ for compound scores; 0.25 for head scores; 0.18 for modifier scores). The difference with respect to the two other ambiguity ranges is the strongest for head scores; it amounts to ≈ 0.20 points. Performance is similar for mid- and high-range ambiguity. For compound and head scores, the mid range performs slightly better than the high range; for modifier scores, it is the reverse. Similarly to low productivity, the better results for low ambiguity suggest that limited semantic dispersion across the occurrences of a given word makes it easier for BERT to learn its meaning. This is in turn beneficial for derived representations of more complex linguistic structures such as compounds.

Overall, these results have shown that compound properties affect predictions of the degree of compositionality, with better performance in the lower ranges of all three properties. This directly echoes the results for type-level word embeddings reported by Schulte im Walde et al. (2016), who similarly suggested that prediction performance is related to broader effects of compound properties on the quality of the underlying meaning representations. Moreover, the prediction of compound and head scores generally follows the same pattern across the three empirical features. Although modifier scores do not align as closely, they too point to an effect of head properties. These results further underscore the importance of the representations of compound heads for the modeling approach we adopted.

6 Conclusion

We have presented an experiment on predicting the degree of compositionality of 280 English noun compounds using a wide range of variants of semantic knowledge derived from pretrained BERT. We have identified a competitive best implementation achieving $\rho = 0.706$ with human judgement and highlighted clear takeaways.

In terms of preprocessing, stronger results were obtained when modeling a larger number of occurrences per compound, whereas controlling for sequence length did not have a clear effect. On embedding computation, different pooling functions led to comparable performance, but there was a clear advantage for layers in the low-to-mid range, which strongly improved on layer combinations used in earlier studies. As for compositionality estimates, it was clearly beneficial to compute them on the token rather than type level, as well as to use (i) representations targeting the constituent of interest; (ii) representations of the surrounding context; (iii) comparisons across complementary – rather than redundant – representational information.

Looking at empirical properties of compounds, low-frequency, low-productivity, and low-ambiguity heads obtain better compositionality predictions. This trend confirms that more limited semantic dispersion makes it easier to model compound meaning. The fact that it holds across compound, head, and modifier compositionality scores highlights the importance of the head in the linguistic structure – and computational modeling – of compounds. Taken together, our results point to important practical decisions when running similar implementations and contribute to our understanding of the way in which BERT represents lexical meaning, supporting the view that the pretrained model encodes at least some aspects of compound semantics.